Problem 7.8

The Kelly family buys a new house for \$93,500 on May 1, 1996. How much was this house worth on May 1, 1992 if real estate prices have risen at a compound rate of 8% per year during that period?

Solution

From May 1, 1992 to May 1, 1996 is 4 years. Let x be the value of the house on May 1, 1992.

Since prices rise at a compound rate of 8% per year,

$$93{,}500 = x(1.08)^4.$$

Therefore,

$$x = \frac{93{,}500}{(1.08)^4} = 93{,}500(1.08)^{-4}.$$

Computing this value,

$$x \approx 68{,}725.29.$$

So the house was worth

$$\boxed{\$68{,}725.29}$$

on May 1, 1992.